Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-7 — Geometric Sequences and Series

# Project Task — Jeddah Greening Programme, Phase 2

*A 20-minute group project. Work in fours. Every member signs the sheet.*

| | |
|---|---|
| Student name : ..... | Date : ..... |
| Group members : ..... ..... ..... ..... | |

**MATERIALS** This sheet · Squared paper · A pencil and ruler · A calculator

## THE DRIVING QUESTION

**Two Jeddah neighbourhoods plant trees in different patterns. Which one has planted more in total, when does the answer change, and what will it cost?**

## THE SITUATION

Vision 2030 sets targets for greening Saudi cities. These are model values for practice, not real data.

Months are counted from month 1. Neighbourhood A doubles its planting each month. Neighbourhood B starts higher but plants only 0.8 times as many as the month before. Each sapling costs a model 25 SAR.

$$A:\ a_1 = 64,\quad r = 2\quad (\text{trees planted in month } n)$$

$$B:\ b_1 = 625,\quad r = 0.8\quad (\text{trees planted in month } n)$$

$$\text{target: } 5\,000 \text{ trees in total}$$

| Month | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| A planted | | | | | |
| B planted | | | | | |

## YOUR TASK 5 steps · 20 minutes

**1 FILL THE TABLE (4 min)**
Write the explicit formula for each neighbourhood, complete both rows of the table for months 1 to 5, and show one calculation for each.

**2 TOTALS (5 min)**
Use the sum formula to find the total planted in the first 5 months by each neighbourhood. Check Neighbourhood B by adding its five terms. Which total is larger?

**3 WHEN DOES IT CHANGE? (4 min)**
Find each total after 6 months. In which month does the total for Neighbourhood A first pass the total for Neighbourhood B?

**4 THE TARGET AND THE COST (4 min)**
Find the first month in which Neighbourhood A has planted more than 5 000 trees in total, and the cost in SAR of all the trees planted up to that month.

**5 PRESENT (3 min)**
One sentence for the planners: which neighbourhood to extend first, and how you know your totals are valid.

## YOUR WORKING write on this sheet

**Step 1 — the formulas, the table and one calculation each:**

**Step 2 — the two totals and the check by adding:**

**Step 3 — the totals after 6 months and the month it changes:**

**Step 4 — the target month and the cost:**

**Step 5 — our sentence:**

**ROLES**

- Reader — states the two patterns and the target
- Calculator — fills the table and the totals
- Checker — adds the terms to test each sum
- Presenter — says the sentence

**DONE WHEN ...**

- Every sum is checked by adding the terms, or by a second method.
- The first term, ratio and number of terms are named before each formula is used.
- The cost is written with units (SAR).
- Your sentence makes sense to someone who does not study maths.